Brandon Liu

U.S. Citizen | brandon-liu.github.io | linkedin.com/in/brandonliu05 | liu3681@purdue.edu | (508) 395-9861

Education

Purdue University — **West Lafayette, IN**
B.S in Mechanical Engineering, Minor in Statistics — *Aug 2023 - May 2027*

Professional Experience

agrobotics — **Salinas, CA**
Mechatronics Engineer Intern — *Jan 2026 - Aug 2026*

- Designed, built, and deployed an in-house 7-DOF robotic arm with COTS BLDC harmonic actuators and CNC-machined 6061 aluminum, increasing reach from 650 to approximately 800 mm; deployed on six data collection units and two prototype robots.
- Compared arm URDF models in MATLAB, demonstrating 63% higher average Yoshikawa manipulability in simulation within the required workspace versus the previous OpenArm v1.
- Redesigned drive units to consolidate parts, reduce material use, and eliminate internal rib alignment and welding; reduced per-unit manufacturing cost by 31% and assembly time by 33% (3 to 2 hours).
- Added two-axis articulation, spring-damper elements, and ballast to front and rear drive units, increasing the field-tested chassis tipping angle from 10° to 25°; sealed ingress points with O-rings and gaskets and tested units in muddy fields.
- Built six teleoperation prototypes for vision-language-action (VLA) policy training, owning arm and chassis design and selecting Jetson Orin electronics; supported approximately 1,200 collected episodes per robot daily and 90% fleet uptime over one month of 16-hour/day, seven-day/week operation.
- Built and tested CAN and RS-485 harnesses for Orin prototypes; selected and began integrating a waterproof Jetson Thor platform to consolidate CAN, Wi-Fi 7, and GNSS hardware and reduce DC-DC converters through native 36 V operation.

Purdue ACM SIGBots — **West Lafayette, IN**
Lead Design Engineer — *Aug 2023 - Present*

- Lead design engineer to 2x World Championship winning team (1/350), prototyping, testing, and fabricating 22 robot subsystems
- Led 50+ mechanical members, assigning subsystem ownership, setting deadlines, and coordinating parts procurement; reviewed and approved mechanism concepts and CAD for tolerances, manufacturability, and packaging.
- Developed Autodesk Inventor lessons and a two-week robot-building qualification challenge to teach recruits CAD and provide hands-on design and fabrication experiences.
- Coordinated field setup and volunteers for 3 robotics competitions for 150+ teams, collectively generating $15,000+ in funding.

ASME Energy, Fuel Cell — **West Lafayette, IN**
Mechanical Engineer — *Aug 2025 - Jan 2026*

- Reviewed component choices for a hydrogen fuel cell system targeting 200 W at 9 V, guiding members in material and manufacturing process selection.
- Presented project design to faculty and ASME board members securing $5,000 in funding.

Semiconductor Packaging Lab, Purdue University — **West Lafayette, IN**
Undergraduate Research Fellow, Instructor: Dr. Tiwei Wei — *Aug 2024 - Dec 2024*

- Extended published MATLAB fluid topology optimization code from 2D to 3D using the Method of Moving Asymptotes (MMA), reducing simulated refrigerant manifold pressure drop by 5% versus the unoptimized geometry for enterprise CPU/GPU cooling.
- Designed manifold prototypes in SolidWorks and fabricated them using SLA 3D printing; prototypes exhibited at CES 2025.

Featured Projects

Competitive Robotics [Portfolio] — *Aug 2023 - Present*
Awards: 2x World Champion, 5x World Championship Awards, 20+ Regional Awards

- **2026-27 Override Robot:** Designed drivetrain, intake, end effector, and a dual-motor, belt-driven differential lift for translation and arm rotation; completed CAD and initial 3D-printed prototypes.
- **2025-26 Push Back Robot:** Designed a linkage-driven slingshot scorer, ground intake, and differential-drive chassis with odometry dead wheels using laser-cut sheet metal, Delrin, and 3D-printed parts.
- **2024-25 High Stakes Robot:** Designed a string-driven cascade lift with COTS slides, climbing hooks, and a two-bar scoring arm and end effector for ladder climbing and elevated ring scoring.

Project Axiom – Adaptive Localization System — *Jun 2026 - Present*

- Defined requirements for sensors, integration across robot platforms, electrostatic discharge protection, and collision resistance.
- Planned calibration-stand testing to characterize drift and estimate measurement variances for Kalman filter tuning.
- Designing a calibration test rig for sensor calibration and data collection

Technical Skills

CAD & Analysis: SolidWorks, NX, Inventor, Simulink, Fusion 360, Onshape, LTSpice
Manufacturing: Design for Manufacturing (DFM), GD&T, 3D Printing, CNC Routing, Laser Cutting, Molding
Programaming/Robotics: MATLAB, R, C, Python, Java, C++, ROS2, CANBus, RS485, PID, Nvidia Jetson